

AIM 5014 Special Topics: AI Safety & Security

Katz School of Science
& Health, MS Program in
Artificial Intelligence

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Instructor Info —



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Course site



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Course Info —



Wednesdays



03:15PM-05:15PM



TBD

Overview

This course provides a comprehensive introduction to the security challenges of modern AI systems. We examine how vulnerabilities can be introduced during system architecture design, model development, training, and deployment and how bad actors can exploit these vulnerabilities via prompt injection, adversarial inputs, data poisoning, and model extraction. Students will also learn about common AI misapplications and the risks introduced by multi-agent collaboration. Alongside these threats, students will examine emerging defenses such as secure architectures, verifiable training, fingerprinting, and prompt-level protections, gaining a deeper understanding of how to assess and improve AI system security. By analyzing real-world breaches and misapplications and engaging in hands-on exercises using open-source AI threat identification and remediation libraries, students will gain valuable insights into the limitations of current AI systems and emerge from the course equipped with the knowledge and skills necessary to build more secure, robust, and trustworthy AI applications.

Material

Required Texts (both available for free online)

- Daniel Jurafsky and James H. Martin, 2026: *Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition, with Language Models*
- Haggai Roitman, 2026: *The Hitchhiker's Guide to Agentic AI: From Foundations to Systems*

Other

Any required papers and book chapters will be provided in the syllabus

Coursework

Students will complete 5 programming assignments, each related to some aspect of the lecture and reading material. The programming assignments should be completed as Jupyter notebooks and uploaded to Canvas.

The final group project can be completed singly or as a group (max 5 members). The project should consist of an application that implements one or more of the concepts described in class. The application will be demoed on the last day of class as a 5-10 minute presentation.

- 10% of the student's grade will be determined by attendance and participation in class
- 70% of the student's grade will be determined by 5 take-home assignments.
- 20% of the student's grade will be determined by a final group project.

Class Schedule

MODULE 1: Introduction to Large Language Models and Agentic AI

Sept 2	Course Introduction & Review of Deep Learning • Overview of AI Safety • Mitre taxonomy • Neural Networks • Deep Learning Lecture notebook: TBD	<u>Required Reading/Viewing</u> • Amodei et al., 2016: Concrete Problems in AI Safety • Goodfellow, Bengio, & Courville: Introduction • Jurafsky & Martin Chap 6: Neural Networks <u>Optional Reading/Viewing</u> • Mitre Atlas • But what is a neural network? • Gradient descent, how neural networks learn • Backpropagation, intuitively.
Sept 9	Transformers & Large Language Models • Attention • Transformers • Large Language Models • Causal language modeling	<u>Required Reading/Viewing</u> • Jurafsky & Martin Chapter 7: Transformers & Pretraining <u>Optional Reading/Viewing</u> • Transformer Explainer • Causal language modeling (HuggingFace tutorial) • Intro to Large Language Models: Karpathy, 2024 • Attention Is All You Need: Vaswani et al., 2017 • The Annotated Transformer
Sept 16	Aligning LLMs to Human Preferences • Reinforcement Learning from Human Feedback • Instruction Tuning • Direct Preference Optimization	<u>Required Reading/Viewing</u> • Jurafsky & Martin Chapter 8: Post-training • Ouyang et al., 2022: Training language models to follow instructions with human feedback • Rafailov et al., 2023: Direct Preference Optimization: Your Language Model is Secretly a Reward Model • Direct Preference Optimization (DPO): A Simplified Explanation <u>Optional Reading/Viewing</u> • Sharma et al., 2024: Towards Understanding Sympathy in Language Models • Ivison et al., 2024: Unpacking DPO and PPO: Disentangling Best Practices for Learning from Preference Feedback • Wen et al., 2024: Language Models Learn to Mislead Humans via RLHF • Qi et al., 2025: Safety Alignment Should be Made More Than Just a Few Tokens Deep
Sept 23	Agentic AI • Retrieval Augmented Generation • Agentic AI – Memory management – Context management and orchestration – Multi-agent architectures	<u>Required Reading/Viewing</u> • Jurafsky & Martin Chap. 11: Information Retrieval and Retrieval-Augmented Generation • Roitman Chap 15: Introduction to Agentic AI • Roitman Chap 17: Agentic Memory Systems, sec 17.1-17.3 • Roitman Chap 18: Agent Harness – Context Management and Orchestration, sec 18.1-18.5 • Roitman Chap 21: Agent-to-Agent Communication (A2A), sec 21.1-21.4 <u>Optional Reading/Viewing</u> • HuggingFace Agents course unit 1: Introduction • Roitman Chap 20: Agent Design Patterns • Roitman Chap 21: Agentic Environments and Benchmarks

MODULE 2: Interpretability & Privacy Preservation

Sept 30

Interpretability

- Circuits
- Mechanistic interpretability
- Sparse autoencoders
- Training data provenance

Required Reading/Viewing

- Rauker, Ho, Casper et al., 2023: SoK: Toward Transparent AI: A Survey on Interpreting Inner Structures
- Bereska & Gavves, 2024: Mechanistic Interpretability for AI Safety - A Review
- Olah et al., 2020: Zoom In: An Introduction to Circuits
- Cunningham et al., 2024: Sparse autoencoders find highly interpretable features in language models
- Grosse et al., 2023: Studying Large Language Model Generalization with Influence Functions

Optional Reading/Viewing

- Bills et al., 2023: Language models can explain neurons in language models
- Li et al., 2025: Training Language Models to Explain Their Own Computations
- Turner et al., 2023: Steering Language Models With Activation Engineering
- Turpin et al., 2023: Language Models Don't Always Say What They Think: Unfaithful Explanations
- Bricken et al., 2023: Towards Monosemanticity: Decomposing Language Models With Dictionary Learning
- Language Models Represent Space and Time – Gournee & Tegmark, 2024
- Turner et al., 2023: Activation addition: Steering language models without optimization

Oct 7

Privacy-Preservation Techniques

- Differential privacy for LLMs
- Federated Learning with LLMs
- LoRA-based finetuning for privacy
- Privacy-preserving data transformation
- Embedding reversal attacks
- Embedding-reversal-resistant techniques

Required Reading/Viewing

- Du et al. 2025: Can Differentially Private Fine-Tuning LLMs Protect Against Privacy Attacks?
- Du et al. 2026: Differentially Private Subspace Fine-Tuning for Large Language Models.
- Yao et al. 2024: Federated Large Language Models: Current Progress and Future Directions.
- Morris et al. 2023: Text Embeddings Reveal (Almost) As Much As Text: Vec2Text Inversion Attacks.
- Zhang et al. 2026: Towards Secure Retrieval-Augmented Generation: Latent Space Exploitation and Embedding Reversal Defenses.
- Gill et al., 2026: BinaryShield: Cross-Service Threat Intelligence in LLM Services using Privacy-Preserving Fingerprints

Optional Reading/Viewing

- Google Research 2025: Fine-tuning LLMs with user-level differential privacy
 - Zhao et al. 2023: Differentially Private Low-Rank Adaptation of Large Language Model Using Federated Learning.
 - Huang et al. 2024: Privacy in Fine-Tuning Large Language Models: Attacks, Defenses, and Future Directions.
 - Tsai et al. 2024: Differentially Private Fine-Tuning of Diffusion Models via LoRA.
 - Song et al. 2020: Information Leakage in Embedding Models: Inversion Attacks and Adversarial Defenses.
 - Security of AI Embeddings Explained
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MODULE 3: Attacks

Oct 14	LLM Prompt Injection (Direct & Indirect) • <i>Direct Prompt Injection / Jailbreaking</i>: Adversaries craft inputs directly targeting the LLM to bypass safety guardrails and override system instructions. • <i>Indirect Prompt Injection</i>: Adversaries embed instructions into unstructured data (e.g., websites, emails) that an AI agent or RAG system ingests during normal processing. • <i>Prompt-Based Data Exfiltration</i>: Manipulating an LLM to covertly embed sensitive session data or credentials into external queries, markdown images, or outbound links. • <i>Multimodal Prompt Injection</i>: Hiding malicious text prompts inside images, audio files, or PDF documents analyzed by vision/multimodal models.	<u>Required Reading/Viewing</u> • EchoLeak: Zero-Click Prompt Injection in M365 Copilot (Zenity Labs) • Bing Chat: Data Exfiltration Exploit Explained(Embrace The Red) • Hacking Google Bard - From Prompt Injection to Data Exfiltration (Embrace The Red) • Universal and Transferable Adversarial Attacks on Aligned Language Models (Zou et al., 2023) • Prompt Injection Attacks Against Vision-Language Models in Oncology (Clusmann et al., Nature Communications 2025) • Agent Smith: A Single Image Can Jailbreak One Million Multimodal LLM Agents (Gu et al., 2024) <u>Optional Reading/Viewing</u> • Hubinger et al., 2024: Sleeper Agents: Training Deceptive LLMs That Persist Through Safety Training. • Wei et al., 2023: Jailbroken: How Does LLM Safety Training Fail? • JailbreakBench: An Open Robustness Benchmark for Jailbreaking Large Language Models (Chao et al., 2024) • Tree of Attacks: Jailbreaking Black-Box Large Language Models Automatically (Mehrotra et al., 2024) • Not What You've Signed Up For: Compromising Real-World LLM-Integrated Applications with Indirect Prompt Injection (Greshake et al., 2023) • TopicAttack: An Indirect Prompt Injection Attack via Topic Transition (Chen et al., EMNLP 2025) • Abusing Images and Audio for Indirect Prompt Injection in Multimodal LLMs (Bagdasaryan et al., 2023) • Visual Adversarial Examples Jailbreak Large Vision-Language Models (Qi et al., 2023)
Oct 21	AI Supply Chain Compromise & Poisoned Tools • <i>Poisoned AI Agent Tools / Skills</i>: Publishing malicious agent extensions or tools on public registries containing hidden prompt injections or system commands. • <i>Package Hallucination Exploitation (Slopsquatting)</i>: Registering malicious software packages under fake package names commonly hallucinated by coding assistants. • <i>Compromised ML Dependencies</i>: Inserting backdoors or malicious code into open-source machine learning libraries or software packages. • <i>Trojanized Pre-Trained Models</i>: Distributing base models that function normally on standard evaluations but exhibit unintended or malicious behavior under specific trigger conditions.	<u>Required Reading/Viewing</u> • Hacking Clawdbot and Eating Lobster Souls • Eating lobster souls Part II: the supply chain • PyTorch Dependency Confusion Attack: Compromised PyTorch-Nightly Dependency (PyTorch Blog) • AI Package Hallucination / Slopsquatting: Slopsquatting: When AI Agents Hallucinate Malicious Packages (Endor Labs) <u>Optional Reading/Viewing</u> • Supply-Chain Poisoning Attacks Against LLM Coding Agent Skill Ecosystems (arXiv 2026) • Asleep at the Keyboard? Assessing the Security of GitHub Copilot's Code Contributions (Pearce et al., 2021) • We Have a Package for You! A Large Scale Study on Package Hallucination in LLMs (arxiv 2024) • Slopsquatting: Exploiting Package Hallucinations in AI-Assisted Workflows (2025) • Poisoning Web-Scale Training Datasets is Practical (Carlini et al., 2024) • Malicious Models on Hugging Face: Silent Trojanized Weights (ReversingLabs, 2024)

Oct 28	AI Infrastructure & Compute Exploitation • <i>Unauthenticated AI Compute Hijacking</i>: Exploiting exposed distributed execution clusters (e.g., Ray, vLLM) to run unauthorized workloads or mine cryptocurrency. • <i>Exposed Model Registries & API Keys</i>: Scanning public registries, vector databases, or storage buckets for exposed API credentials or open model management interfaces. • <i>Orchestration Framework Exploitation</i>: Targeting vulnerabilities in tools like LangChain, LlamaIndex, or Kubeflow to achieve local or remote arbitrary code execution. • <i>Vector Database Exploitation</i>: Unauthorized manipulation or exfiltration of embeddings and metadata stored in vector search engines due to permissive access control policies.	<u>Required Reading/Viewing</u> • ShadowRay: First Known Attack Campaign Targeting AI Workloads Actively Exploited In The Wild (Oligo Security) • LLMjacking evolved: Attackers are using stolen AI compute to build offensive agentic tools • Silent Sabotage: Weaponizing AI Models in Exposed Containers (Trend Micro) • Truffle Security 2025: 12,000 Live API Keys and Passwords Exposed in Public Web Datasets Ingested by LLMs. <u>Optional Reading/Viewing</u> • Red Hat Security 2025: LangChain Serialization Vulnerability Allowing Arbitrary Code Execution (CVE-2025-68664). • Gourion et al. 2025: ShadowMQ: How Code Reuse Spread Critical Vulnerabilities Across the AI Ecosystem. • OWASP GenAI Committee 2025: LLM08: Vector and Embedding Weaknesses in RAG Architectures.
Nov 4	Model Inversion & IP Theft (Model Extraction) • <i>Black-Box Model Theft via API Queries</i>: Querying an API with carefully crafted inputs to reverse-engineer model parameters and create a duplicate model. • <i>Training Data Extraction</i>: Reconstructing verbatim memorized training samples (including PII, passwords, or copyrighted text) through automated generation prompts. • <i>System Prompt & Meta-Prompt Extraction</i>: Extracting hidden system instructions, guardrail logic, or fine-tuning guidelines by tricking the LLM into repeating its initial prompt context.	<u>Required Reading/Viewing</u> • Nasr et al. 2023: Scalable Extraction of Training Data from Production Language Models. • Carlini et al. 2024: Stealing Part of a Production Model • LouisShark et al. 2024: Collection of Extracted System Prompts and Meta-Prompts. <u>Optional Reading/Viewing</u> • SentinelOne: Model Inversion Attacks: Risks and Defenses Explained (blog post) • Geiping et al. 2024: Coercing LLMs to do and reveal (almost) anything. • Chen et al. 2025: Exploiting Prompt-Induced Confidence for Black-Box Attacks on LLMs.
Nov 11	AI Agent Misuse & Unauthorized Agency • <i>Privilege Escalation via Tool Binding</i>: Exploiting over-privileged tools assigned to an AI agent to read/write files or execute commands beyond user authorization. • <i>Unauthorized Web Browsing & API Actions</i>: Tricking an active web agent into performing actions on behalf of a logged-in user, such as sending emails or transferring funds. • <i>AI Agent Configuration Modification</i>: Altering systemic instructions or tool parameter definitions stored in agent state configs to maintain persistence or subvert safety rules.	<u>Required Reading/Viewing</u> • Microsoft Copilot Studio Vulnerabilities: Explained (Zenity Labs) • MITRE ATLAS Investigation into OpenClaw Agent Compromise (MITRE PDF) • AI with Aish: OpenClaw Explained: Open-Source Control Layer for Agents (blog post) • The Hidden Threat: How Indirect Prompt Injections Exploit AI Agents and Why Concealment Matters (ARSA Technology) <u>Optional Reading/Viewing</u> • DeBeneddetti 2024: AgentDojo: A Dynamic Environment to Evaluate Prompt Injection Attacks and Defenses for LLM Agents • Andriushchenko et al., 2024: AgentHarm: A Benchmark for Measuring Harmfulness of LLM Agents • Chen et al., 2026: IterInject: Indirect Prompt Injection Against LLM Agents via Feedback-Guided Iterative Optimization

MODULE 4: Defenses

Nov 18

Runtime Defenses

- Architectural controls
- Runtime guardrail layers

Required Reading/Viewing

- Andersen et al. 2025: AgentHarm: Benchmarking Robustness of LLM Agents on Harmful Tasks.
- Inan et al. 2024: Llama Guard 3: Vision and Text Safeguards for Human-AI Conversations.
- TrueFoundry Engineering 2026: Prompt Injection Defense at the AI Gateway.

Optional Reading/Viewing

- Zhou et al. 2024: AutoDefense: Multi-Agent LLM Defense against Jailbreak Attacks.
- Digital Applied 2026: LLM Guardrails: Production Safety Layers Reference.

Nov 25

Offline Defenses

- Alignment
- Pre-training mitigations
- LLM Unlearning
- Refusal training

Required Reading/Viewing

- O'Brien et al. 2025: Deep Ignorance: Filtering Pretraining Data Builds Tamper-Resistant Safeguards.
- OpenAI, 2022: DALL·E 2 pre-training mitigations
- Yao et al. 2024: Large Language Model Unlearning
- Birhane et al., 2023: On Hate Scaling Laws For Data-Swamps
- Eldan & Rusinovich, 2023: Who's Harry Potter? Approximate Unlearning for LLMs
- Bai et al., 2022: Training a Helpful and Harmless Assistant with Reinforcement Learning from Human Feedback
- Tice et al., 2026: Alignment Pretraining: AI Discourse Causes Self-Fulfilling (Mis)alignment

Optional Reading/Viewing

- Guo et al. , 2026: LLMs Can Unlearn Refusal with Only 1,000 Benign Samples
 - Qi et al. 2025: Safety Alignment Should be Made More Than Just a Few Tokens Deep.
 - Cooper et al., 2024: Machine Unlearning Doesn't Do What You Think: Lessons for Generative AI Policy
 - Eldan & Li, 2023: TinyStories: How Small Can Language Models Be and Still Speak Coherent English?
 - Gunasekar et al., 2023: Textbooks Are All You Need
 - Jian et al., 2024: Investigating Data Contamination for Pre-training Language Models
 - Liu et al., 2025: Language Models May Verbatim Complete Text They Were Not Explicitly Trained On
 - Schaeffer et al., 2025: Position: Model Collapse Does Not Mean What You Think
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Dec 2	AI Supervision & Governance • Scalable AI oversight & governance • AI risk frameworks • Benchmarking AGI • AI deception prevention	<u>Required Reading/Viewing</u> • Irving & Amodei, 2018: AI Safety via Debate (Blog post) • Burns et al., 2023: Weak-to-strong generalization (Blog post) • Bengio et al., 2024: Managing extreme AI risks amid rapid progress • Ngo et al., 2024: The Alignment Problem from a Deep Learning Perspective: A Position Paper <u>Optional Reading/Viewing</u> • Kasirzadeh, 2023: Two Types of AI Existential Risk: Decisive and Accumulative • Alexander, 2022: Biological Anchors: A Trick That Might Or Might Not Work • Bowman et al., 2022: Measuring Progress on Scalable Oversight for Large Language Models • Hendrycks et al., 2023: An Overview of Catastrophic AI Risks • Morris et al., 2024: Position: Levels of AGI for Operationalizing Progress on the Path to AGI • ELK Summary (Hobbhahn, 2022: Full report)
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Dec 9 Final Presentations